ALICE L WALTON SCHOOL OF MEDICINE

Spectrum of Activity and indications–2 β-lactams and 30s Inhibitors

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Learning Objectives (1)

After completing this brick, you will be able to:

1. Describe the spectrum of activity of β-lactams, Aminoglycosides, and Tetracycline.

CASE CONNECTION

AL, a 16-year-old boy presents to the clinic with a three-day history of redness, swelling, and warmth on his right lower leg. He reports that the area has been gradually increasing in size and is tender to touch. There is no drainage or pus from the affected area, and he denies any recent trauma or insect bites. The boy has a low-grade fever and feels generally unwell. On examination, the skin on his right lower leg is erythematous, edematous, and warm, with well-demarcated borders. The clinical presentation is consistent with non-purulent cellulitis, likely caused by a bacterial infection.

What can you give AL to treat his condition? Consider your answer as you read, and we'll revisit AL at the end of the brick.

GO TO CONCLUSION ↓

In the previous spectrum of activity and indications brick, we discussed gram-positive coverage for the β-lactams and vancomycin. Since we covered some important gram-negative organisms this week, let's first look at gram-negative coverage before we proceed to other antibiotics.

Review: What gram-negative organisms does Vancomycin cover?

What gram-negative organisms can be treated with the β-lactams? (LO#1)

Amoxicillin and Ampicillin

Aminopenicillins such as **Amoxicillin and Ampicillin**, in combination with b-lactamase inhibitors, provide good gram-negative coverage and coverage of many atypicals. Ampicillin (with sulbactam) is effective against *Neisseria meningitidis* which can cause bacterial meningitis and septicemia. They do not cover the Extended Spectrum beta-lactamase producing Enterobacteriaceae bacteria or Pseudomonas. Recall from the prior brick that they provide good Gram-Positive coverage (except MRSA) including Enterococcus.

CLINICAL CORRELATION

Acute bacterial sinusitis, an upper respiratory tract infection, is often caused by *S. pneumoniae, H. influenzae, M. catarrhalis*. Treating acute sinusitis with Amoxicillin/clavulanate is very common.

Piperacillin-tazobact

Remember, we previously discussed that piperacillin-tazobactam has broad gram-positive coverage (except MRSA) and gram-negative coverage. It also covers other gram-negative bacteria, such as Enterobacter species (including *Escherichia coli*), Klebsiella, Serratia, and Proteus. Piperacillin-tazobactam has one of the broadest Gram Negative coverages, including ESBLs and Pseudomonas. You can now see why piperacillin-tazobactam is used for initial empiric therapy.

Carbapenems and Monobactams

The easiest way to remember carbapenem coverage is that - 'it covers all gram-positive and gram-negative bacteria.' **Carbapenems** are a class of β-lactam antibiotics known for their broad-spectrum activity against a wide range of bacteria, including many gram-positive (except MRSA) and gram-negative organisms. Carbapenems are considered the most reliable treatment for infections caused by extended-spectrum beta-lactamase (ESBL)-producing bacteria. The ESBLs produce β-lactamase enzymes, which can degrade other β-lactam antibiotics. Carbapenems broad-spectrum activity and resistance to β-lactamase enzymes make them valuable in treating complicated infections, including intra-abdominal infections, urinary tract infections, and hospital-acquired pneumonia. All of these infections are hard to treat, and carbapenems are always reserved to treat these infections.

Remember that carbapenems (Ertapenem, Meropenem, Doripenem, Imipenem/Cilastatin) can cover most gram-negative bacteria. However, there are some exceptions. The most important **exception** is **Ertapenem.** Ertapenem does not have coverage against Pseudomonas, Enterococci, and Acinetobacter (see mnemonic below).

Mnemonic

Ert**AP**enem does not cover **PEA**

Monobactams are also effective against aerobic gram-negative bacteria and exhibit a high level of resistance to beta-lactamases produced by these organisms. Unlike penicillins, monobactams do not treat Gram Positive bacteria or anaerobes. However, this makes them particularly useful in treating infections caused by gram-negative bacilli without disrupting the patient's microbiota.

What is the primary use of monobactams?

It is important to note that Pencillins and Anti-Staphylococcal penicillins (Nafcillin/Oxacillin) are not effective against many gram-negative bacteria. In most cases, bacterial resistance to these agents have been developed by expression of β-lactamase enzymes and efflux pumps.

What other antibiotic agents have good gram-negative coverage?

This week we discussed the pharmacology of tetracyclines and aminoglycosides, both of which have good gram-negative coverage.

Aminoglycosides

Aminoglycosides are potent antibiotics with a broad spectrum of activity, primarily effective against aerobic gram-negative bacteria. They are particularly effective against members of the Enterobacteriaceae family, including Escherichia coli, Klebsiella pneumoniae, Enterobacter, Providencia, Proteus, Morganella, and Serratia species. Additionally, aminoglycosides are effective against *Pseudomonas aeruginosa*. However, they have limited activity against gram-positive bacteria and are typically used synergistically with other antibiotics for such infections. Aminoglycosides are not effective against anaerobic bacteria or [illegible]

Tetracyclines

Tetracyclines have a broad spectrum of activity against a wide range of microorganisms. They are effective against gram-positive bacteria such as Staphylococcus (including MRSA), Streptococcus, and Enterococcus (including VRE). Doxycycline has wide gram-negative bacteria, and the coverage includes Acinetobacter, Legionella, *Neisseria gonorrhoeae*, and *Haemophilus influenzae*. Tetracycline (Doxycycline) is commonly used to treat respiratory tract infections such as community acquired pneumonia.

CLINICAL CORRELATION

Many skin and soft tissue infections, whether they are purulent or non-purulent, are often treated with doxycycline because of it's broad Gram-positive coverage that includes MRSA.

A commonly used tetracycline (both inpatient and outpatient) is Doxycycline. Use of **Doxycycline** is very common in treating atypical pathogens. They include pathogens such as *Mycoplasma pneumoniae*, *Chlamydia trachomatis*, and *Rickettsiae*. Doxycycline is the first-line treatment for Rickettsiae and Lyme's disease.

What is the first-line treatment for Rickettsiae and Lyme's disease?

CLINICAL CORRELATION

The common bacterial causes are *Streptococcus pneumoniae, Nesseria meningitidis*, and *Haemophilus influenzae*. In adults and patients >50 years, combination of Ceftriaxone with or without Vancomycin provides coverage against these organisms.

In neonates (<1 month) and patients >50 or immunocompromised, *Listeria monocytogenes* infection has a higher occurrence rate. In this case *Ampicillin* can provide effective treatment. *Gentamicin* with other cephalosporins (cefepime, cefotaxime) can also be used for neonates. In patients above 50, *Ampicillin* (for Listeria coverage) with Ceftriaxone and Vancomycin is used for treatment.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Based on your reading, what would you recommend for AL?

A diagnosis of non-p[illegible] we do not know the pathogen that is causing the infection. Typically, these infections are attributed to *S. pyogenes* (Group A *Streptococcus*) and *S. aureus*. These bacteria can be covered using first-generation cephalosporins. In AL's case, oral antibiotics, such as cephalexin can be used. An alternative to cephalexin is oxacillin (anti-staphyloccocal penicillin). Supportive care measures like rest, elevation of the affected leg, and pain management with over-the-counter analgesics will also be helpful. Follow-up is advised to monitor the response to treatment and ensure resolution of the infection. AL should complete the full course of antibiotics and ensure that she is hydrated to avoid reoccurrence.

Summary

- **Aminopenicillins** like Amoxicillin and Ampicillin, combined with β-lactamase inhibitors, provide good gram-negative coverage and Neisseria.
- **Piperacillin-tazobactam** has broad gram-positive (except MRSA) and gram-negative coverage, including *Pseudomonas* and *Campylobacter*, making it suitable for initial empiric therapy.
- **Aminoglycosides** are potent antibiotics effective against aerobic gram-negative bacteria, including Enterobacteriaceae family members and *Pseudomonas aeruginosa,* but have limited activity against gram-positive bacteria and are not effective against anaerobic bacteria or atypical pathogens.
- **Tetracyclines**, including doxycycline, have a broad spectrum of activity against gram-positive bacteria (including MRSA and VRE), gram-negative bacteria, and atypical pathogens, but are not effective against Pseudomonas aer[illegible]ainst anaerobes.

Review Questions

1. Extended spectrum β-lactamase (ESBLs) producing bacteria can be best covered by which of following?

A. ○ Linezolid
B. ○ Meropenem
C. ○ Penicillin
D. ● Vancomycin

Explanation (requires correct answer)

2. Aminoglycosides can be used by itself to cover which one of the following pathogen?

A. ○ *Clostridium difficile*
B. ○ *Streptococcus epidermidis*
C. ○ *Staphylococcus aureus*
D. ○ *Proteus Mirabilis*

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Explanation (requires correct answer)

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☐ **Describe the spectrum of activity of β-lactams, Aminoglycosides, and Tetracycline.**

Objective Confidence: 0/1